

Runbook: xHTTP через CDN с uplink only GET

Чистый сервер: Nginx + Remnawave Node + Xray xHTTP packet-up GET

Перед началом: Документ использует placeholders: ORIGIN_DOMAIN, CDN_DOMAIN, CDN_TECH_DOMAIN, ORIGIN_IP. Заменяй их на свои значения перед выполнением команд.

0. Схема работы

Цель - настроить xHTTP через CDN без POST/PUT, используя только GET uplink. Xray слушает только localhost, публично доступен только Nginx на 443.

```
Client / Happ
-> CDN_DOMAIN:443 или CDN_TECH_DOMAIN:443
-> CDN edge
-> ORIGIN_DOMAIN:443 / ORIGIN_IP:443
-> Nginx
-> /api/v4/media/session/poll
-> 127.0.0.1:11443
-> Remnawave Node / Xray inbound
```

Параметр	Значение
ORIGIN_DOMAIN	Домен origin, например origin.example.com
ORIGIN_IP	IP чистого сервера
CDN_TECH_DOMAIN	Технический домен CDN, например edge.example-cdn.net
Xray local port	11443, только 127.0.0.1
XHTTP path	/api/v4/media/session/poll
uplink method	GET
session placement	cookie media_sid
seq placement	query offset
uplink data placement	header X-Playback-Token

CDN: POST/PUT на CDN не нужны. Кэширование для xHTTP path не включать: в query/cookie/header находятся служебные данные транспорта.

1. DNS и подготовка

До выпуска сертификата origin-домен должен указывать на сервер. CDN-домен или технический домен CDN используется в Host для клиентов.

```
# DNS на время выпуска origin-сертификата
ORIGIN_DOMAIN A ORIGIN_IP

# Проверка
dig +short ORIGIN_DOMAIN @1.1.1.1

auth result expected:
ORIGIN_IP
```

2. Установка пакетов и базовая сеть

```
apt update && apt -y upgrade
apt install -y nginx certbot curl wget git unzip nano httpd jq ufw openssl ca-certificates dnsutils
curl -fsSL https://get.docker.com | sh
systemctl enable --now docker
```

Системные лимиты и TCP-настройки

```
cat >/etc/sysctl.d/99-vpn-connection-limits.conf <<'EOF'
fs.file-max = 2097152
fs.nr_open = 2097152

net.core.default_qdisc = fq
net.ipv4.tcp_congestion_control = bbr
net.ipv4.tcp_fastopen = 3
net.ipv4.tcp_mtu_probing = 1
net.ipv4.tcp_slow_start_after_idle = 0

net.core.somaxconn = 65535
net.ipv4.tcp_max_syn_backlog = 65535
net.ipv4.ip_local_port_range = 1024 65535

net.ipv4.tcp_fin_timeout = 15
net.ipv4.tcp_tw_reuse = 1
net.ipv4.tcp_keepalive_time = 300
net.ipv4.tcp_keepalive_intvl = 30
net.ipv4.tcp_keepalive_probes = 5

net.netfilter.nf_conntrack_max = 262144
net.netfilter.nf_conntrack_tcp_timeout_established = 3600
net.netfilter.nf_conntrack_tcp_timeout_time_wait = 30
net.netfilter.nf_conntrack_tcp_timeout_close_wait = 30
net.netfilter.nf_conntrack_tcp_timeout_fin_wait = 30
EOF

modprobe nf_conntrack || true
echo nf_conntrack >/etc/modules-load.d/nf_conntrack.conf
sysctl --system
```

3. Firewall

Публично нужны только 80/443. Remnawave Node API порт открывай только под IP панели. Клиентский VPN-трафик через этот порт не ходит.

```
ufw default deny incoming
ufw default allow outgoing
ufw allow 22/tcp
ufw allow 80/tcp
ufw allow 443/tcp
# ufw allow from PANEL_IP to any port NODE_API_PORT proto tcp
ufw --force enable
ufw status verbose
```

4. Remnawave Node

Если Node уже подключена к панели, этот раздел пропусти. Если сервер чистый, создай ноду в Remnawave Panel, скопируй docker-compose и запусти его на сервере.

```
mkdir -p /opt/remnanode /var/log/remnanode
cd /opt/remnanode
nano docker-compose.yml
# вставь compose из Remnawave Panel
```

```
docker compose up -d
docker ps
docker logs --tail=100 remnanode
```

Лимиты Node: В docker-compose желательно добавить ulimits nofile 1048576. Проверка: `docker exec remnanode sh -lc "ulimit -n"`.

5. Выпуск SSL для origin

```
export ORIGIN_DOMAIN="ORIGIN_DOMAIN"
export EMAIL="admin@example.com"

systemctl stop nginx || true
certbot certonly --standalone -d "$ORIGIN_DOMAIN" --non-interactive --agree-tos -m "$EMAIL"

ls -la /etc/letsencrypt/live/$ORIGIN_DOMAIN/
systemctl start nginx || true
```

6. Заглушка сайта

```
mkdir -p /var/www/decoy/assets

cat >/var/www/decoy/index.html <<'EOF'
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>Media Session Gateway</title>
  <meta name="viewport" content="width=device-width,initial-scale=1">
  <link rel="stylesheet" href="/assets/app.css">
</head>
<body>
  <main>
    <div class="badge">Media API</div>
    <h1>Media Session Gateway</h1>
    <p>Status: operational</p>
    <p class="muted">This service handles media session polling, delivery checks and application synchronization.</p>
  </main>
  <script src="/assets/app.js"></script>
</body>
</html>
EOF

cat >/var/www/decoy/assets/app.css <<'EOF'
body {
  margin: 0;
  font-family: system-ui, -apple-system, BlinkMacSystemFont, "Segoe UI", sans-serif;
  background: #f5f7fb;
  color: #20242c;
}
main {
  max-width: 760px;
  margin: 12vh auto;
  padding: 34px;
  background: #fff;
  border-radius: 18px;
  box-shadow: 0 18px 55px rgba(0,0,0,.08);
}
.badge {
  display: inline-block;
  padding: 6px 10px;
  border-radius: 999px;
  background: #eef6ff;
  color: #2764a7;
  font-size: 13px;
```

```
}
h1 { margin: 18px 0 10px; }
.muted { color: #667085; }
EOF

cat >/var/www/decoy/assets/app.js <<'EOF'
console.log("Media Session Gateway");
EOF

chown -R www-data:www-data /var/www/decoy
```

7. Полный Nginx

Важно: Перед вставкой замени ORIGIN_DOMAIN в site config на свой домен. Храу должен слушать 127.0.0.1:11443.

Бэкап

```
cp /etc/nginx/nginx.conf /etc/nginx/nginx.conf.bak.$(date +%F-%H%M%S)
mkdir -p /root/nginx-backup-$(date +%F-%H%M%S)
cp -a /etc/nginx/sites-available /etc/nginx/sites-enabled /etc/nginx/conf.d /root/nginx-backup-$(date +%F-%H%M%S)/
2>/dev/null || true
```

/etc/nginx/nginx.conf

```
user www-data;
worker_processes auto;
worker_rlimit_nofile 1048576;
pid /run/nginx.pid;

include /etc/nginx/modules-enabled/*.conf;

events {
    worker_connections 65535;
    multi_accept on;
    use epoll;
}

http {
    sendfile on;
    tcp_nopush on;
    tcp_nodelay on;

    types_hash_max_size 4096;
    server_tokens off;

    include /etc/nginx/mime.types;
    default_type application/octet-stream;

    ssl_protocols TLSv1.2 TLSv1.3;
    ssl_prefer_server_ciphers off;
    ssl_session_cache shared:SSL:20m;
    ssl_session_timeout 1d;

    access_log /var/log/nginx/access.log;
    error_log /var/log/nginx/error.log warn;

    gzip off;

    keepalive_timeout 30s;
    keepalive_requests 1000;

    reset_timedout_connection on;
    send_timeout 60s;
    client_body_timeout 60s;
```

```

client_header_timeout 60s;

client_max_body_size 0;
client_header_buffer_size 16k;
large_client_header_buffers 8 32k;

proxy_headers_hash_max_size 1024;
proxy_headers_hash_bucket_size 128;

map_hash_bucket_size 128;
map_hash_max_size 4096;

log_format xhttp_min '$remote_addr $time_local host="$host" proto="$server_protocol" '
                    '"$request_method $request_uri" st=$status bytes=$body_bytes_sent '
                    'cl="$content_length" req_len=$request_length '
                    'rt=$request_time urt="$upstream_response_time" '
                    'ua="$http_user_agent"';

include /etc/nginx/sites-enabled/*;
}

```

Команда записи nginx.conf

```

cat >/etc/nginx/nginx.conf <<'EOF'
user www-data;
worker_processes auto;
worker_rlimit_nofile 1048576;
pid /run/nginx.pid;

include /etc/nginx/modules-enabled/*.conf;

events {
    worker_connections 65535;
    multi_accept on;
    use epoll;
}

http {
    sendfile on;
    tcp_nopush on;
    tcp_nodelay on;

    types_hash_max_size 4096;
    server_tokens off;

    include /etc/nginx/mime.types;
    default_type application/octet-stream;

    ssl_protocols TLSv1.2 TLSv1.3;
    ssl_prefer_server_ciphers off;
    ssl_session_cache shared:SSL:20m;
    ssl_session_timeout 1d;

    access_log /var/log/nginx/access.log;
    error_log /var/log/nginx/error.log warn;

    gzip off;

    keepalive_timeout 30s;
    keepalive_requests 1000;

    reset_timedout_connection on;
    send_timeout 60s;
    client_body_timeout 60s;
    client_header_timeout 60s;

    client_max_body_size 0;
    client_header_buffer_size 16k;
    large_client_header_buffers 8 32k;
}

```

```

proxy_headers_hash_max_size 1024;
proxy_headers_hash_bucket_size 128;

map_hash_bucket_size 128;
map_hash_max_size 4096;

log_format xhttp_min '$remote_addr $time_local host="$host" proto="$server_protocol" '
                    '$request_method $request_uri' st=$status bytes=$body_bytes_sent '
                    'cl="$content_length" req_len=$request_length '
                    'rt=$request_time urt="$upstream_response_time" '
                    'ua="$http_user_agent"';

include /etc/nginx/sites-enabled/*;
}
EOF

```

/etc/nginx/sites-available/ORIGIN_DOMAIN

```

upstream xray_xhttp_get {
    server 127.0.0.1:11443;
    keepalive 256;
}

server {
    listen 80 default_server;
    listen [::]:80 default_server;

    server_name ORIGIN_DOMAIN;

    location ^~ /.well-known/acme-challenge/ {
        root /var/www/decoy;
    }

    location / {
        return 301 https://$host$request_uri;
    }
}

server {
    listen 443 ssl http2 default_server;
    listen [::]:443 ssl http2 default_server;

    server_name ORIGIN_DOMAIN;

    ssl_certificate      /etc/letsencrypt/live/ORIGIN_DOMAIN/fullchain.pem;
    ssl_certificate_key  /etc/letsencrypt/live/ORIGIN_DOMAIN/privkey.pem;

    server_tokens off;

    root /var/www/decoy;
    index index.html;

    client_max_body_size 0;
    client_body_timeout 900s;
    client_header_timeout 900s;
    client_header_buffer_size 16k;
    large_client_header_buffers 8 32k;

    access_log /var/log/nginx/access.log;
    error_log /var/log/nginx/error.log warn;

    location = /health {
        default_type application/json;
        add_header Cache-Control "no-store" always;
        return 200 '{"status":"ok"}';
    }

    location / {

```

```

    try_files $uri $uri/ /index.html;
    add_header Cache-Control "public, max-age=300" always;
}

location ~* \.(css|js|png|jpg|jpeg|svg|ico|webp|woff2)$ {
    try_files $uri =404;
    expires 1h;
    add_header Cache-Control "public, max-age=3600" always;
}

location ^~ /api/v4/media/session/poll {
    access_log /var/log/nginx/xhttp_access.log xhttp_min;

    proxy_pass http://xray_xhttp_get;
    proxy_http_version 1.1;

    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto https;
    proxy_set_header Connection "";

    proxy_buffering off;
    proxy_request_buffering off;
    proxy_cache off;
    gzip off;

    add_header Cache-Control "no-store" always;
    add_header X-Accel-Buffering "no" always;

    proxy_connect_timeout 30s;
    proxy_read_timeout 900s;
    proxy_send_timeout 900s;
}
}

```

Команда записи site config

```

export ORIGIN_DOMAIN="ORIGIN_DOMAIN"
cat >/etc/nginx/sites-available/$ORIGIN_DOMAIN <<EOF
upstream xray_xhttp_get {
    server 127.0.0.1:11443;
    keepalive 256;
}

server {
    listen 80 default_server;
    listen [::]:80 default_server;

    server_name $ORIGIN_DOMAIN;

    location ^~ /.well-known/acme-challenge/ {
        root /var/www/decoy;
    }

    location / {
        return 301 https://$host$request_uri;
    }
}

server {
    listen 443 ssl http2 default_server;
    listen [::]:443 ssl http2 default_server;

    server_name $ORIGIN_DOMAIN;

    ssl_certificate      /etc/letsencrypt/live/$ORIGIN_DOMAIN/fullchain.pem;
    ssl_certificate_key  /etc/letsencrypt/live/$ORIGIN_DOMAIN/privkey.pem;

    server_tokens off;
}

```

```

root /var/www/decoy;
index index.html;

client_max_body_size 0;
client_body_timeout 900s;
client_header_timeout 900s;
client_header_buffer_size 16k;
large_client_header_buffers 8 32k;

access_log /var/log/nginx/access.log;
error_log /var/log/nginx/error.log warn;

location = /health {
    default_type application/json;
    add_header Cache-Control "no-store" always;
    return 200 '{"status":"ok"}';
}

location / {
    try_files $uri $uri/ /index.html;
    add_header Cache-Control "public, max-age=300" always;
}

location ~* \.(css|js|png|jpg|jpeg|svg|ico|webp|woff2)$ {
    try_files $uri =404;
    expires 1h;
    add_header Cache-Control "public, max-age=3600" always;
}

location ^~ /api/v4/media/session/poll {
    access_log /var/log/nginx/xhttp_access.log xhttp_min;

    proxy_pass http://xray_xhttp_get;
    proxy_http_version 1.1;

    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto https;
    proxy_set_header Connection "";

    proxy_buffering off;
    proxy_request_buffering off;
    proxy_cache off;
    gzip off;

    add_header Cache-Control "no-store" always;
    add_header X-Accel-Buffering "no" always;

    proxy_connect_timeout 30s;
    proxy_read_timeout 900s;
    proxy_send_timeout 900s;
}
}
EOF

rm -f /etc/nginx/sites-enabled/default
ln -sf /etc/nginx/sites-available/$ORIGIN_DOMAIN /etc/nginx/sites-enabled/$ORIGIN_DOMAIN

mkdir -p /etc/systemd/system/nginx.service.d
cat >/etc/systemd/system/nginx.service.d/limits.conf <<'EOF'
[Service]
LimitNOFILE=1048576
TasksMax=infinity
EOF

systemctl daemon-reload
nginx -t && systemctl restart nginx
systemctl status nginx --no-pager

```


8. Remnawave Config Profile

Вставь этот JSON в Config Profiles. Tag можно оставить как есть или заменить на уже используемый tag, если не хочешь менять привязки Host/Squad.

Полный Config Profile

```
{
  "log": {
    "loglevel": "warning"
  },
  "dns": {
    "servers": [
      {
        "address": "https://dns.google/dns-query",
        "skipFallback": false
      }
    ],
    "queryStrategy": "UseIPv4"
  },
  "inbounds": [
    {
      "tag": "cdn-get-inbound",
      "port": 11443,
      "listen": "127.0.0.1",
      "protocol": "vless",
      "settings": {
        "clients": [],
        "decryption": "none"
      },
      "sniffing": {
        "enabled": true,
        "destOverride": [
          "http",
          "tls",
          "quic"
        ]
      }
    },
    {
      "streamSettings": {
        "network": "xhttp",
        "security": "none",
        "xhttpSettings": {
          "mode": "packet-up",
          "path": "/api/v4/media/session/poll",
          "extra": {
            "headers": {
              "Accept": "application/vnd.api+json, application/json, text/plain, */*",
              "Accept-Language": "ru-RU, ru;q=0.9, en-US;q=0.8, en;q=0.7",
              "Cache-Control": "no-cache",
              "Pragma": "no-cache"
            }
          },
          "xPaddingBytes": "48-320",
          "xPaddingObfsMode": true,
          "xPaddingMethod": "tokenish",
          "xPaddingPlacement": "queryInHeader",
          "xPaddingHeader": "X-Rewrite-URL",
          "xPaddingKey": "q",
          "sessionPlacement": "cookie",
          "sessionKey": "media_sid",
          "seqPlacement": "query",
          "seqKey": "offset",
          "uplinkHTTPMethod": "GET",
          "uplinkDataPlacement": "header",
          "uplinkDataKey": "X-Playback-Token",
          "serverMaxHeaderBytes": 32768,
          "scMaxEachPostBytes": "1536-6144",
          "scMinPostsIntervalMs": "4-18",
          "scMaxBufferedPosts": 32,
          "xmux": {
            "maxConcurrency": "6-16",
```

```
        "cMaxReuseTimes": "36-96",
        "hMaxRequestTimes": "320-640",
        "hMaxReusableSecs": "720-1800",
        "hKeepAlivePeriod": 0
    }
}
}
},
],
"outbounds": [
    {
        "tag": "DIRECT",
        "protocol": "freedom"
    },
    {
        "tag": "BLOCK",
        "protocol": "blackhole"
    }
],
"routing": {
    "rules": [
        {
            "type": "field",
            "ip": [
                "geoip:private"
            ],
            "outboundTag": "BLOCK"
        },
        {
            "type": "field",
            "protocol": [
                "bittorrent"
            ],
            "outboundTag": "BLOCK"
        }
    ]
}
}
```

9. Host в Remnawave

Создай Host на inbound из Config Profile. Для технического CDN-домена:

Параметр	Значение
Remark	CDN GET xHTTP
Inbound	cdn-get-inbound или твой tag
Address	CDN_TECH_DOMAIN
Port	443
SNI	CDN_TECH_DOMAIN
Host	CDN_TECH_DOMAIN
Path	/api/v4/media/session/poll
Security Layer	TLS
Network	xHTTP
Mode	packet-up
ALPN	h2,http/1.1
Fingerprint	chrome

xHTTP extra params для Host - без поля path

```
{
  "headers": {
    "Accept": "application/vnd.api+json, application/json, text/plain, */*",
    "Accept-Language": "ru-RU,ru;q=0.9,en-US;q=0.8,en;q=0.7",
    "Cache-Control": "no-cache",
    "Pragma": "no-cache"
  },
  "xPaddingBytes": "48-320",
}
```

```

"xPaddingObfsMode": true,
"xPaddingMethod": "tokenish",
"xPaddingPlacement": "queryInHeader",
"xPaddingHeader": "X-Rewrite-URL",
"xPaddingKey": "q",
"sessionPlacement": "cookie",
"sessionKey": "media_sid",
"seqPlacement": "query",
"seqKey": "offset",
"uplinkHTTPMethod": "GET",
"uplinkDataPlacement": "header",
"uplinkDataKey": "X-Playback-Token",
"serverMaxHeaderBytes": 32768,
"scMaxEachPostBytes": "1536-6144",
"scMinPostsIntervalMs": "4-18",
"scMaxBufferedPosts": 32,
"xmux": {
  "maxConcurrency": "6-16",
  "cMaxReuseTimes": "36-96",
  "hMaxRequestTimes": "320-640",
  "hMaxReusableSecs": "720-1800",
  "hKeepAlivePeriod": 0
}
}

```

10. CDN settings

Параметр	Значение
Origin	ORIGIN_DOMAIN:443 или ORIGIN_IP:443 с Host/SNI = ORIGIN_DOMAIN
Protocol to origin	HTTPS
Methods	GET/HEAD/OPTIONS. POST/PUT можно выключить
Cache	Off для VPN/xHTTP path
Query string	Не резать
Cookies	Не резать
Headers	Не резать, особенно X-Playback-Token и X-Rewrite-URL
Gzip/Brotli	Off
HTTP/3	Off для первого запуска
Secure token / WAF / CORS	Off

Кэш: Если CDN позволяет path rules, можно кэшировать только статические файлы деску-сайта. Но /api/v4/media/session/poll всегда должен быть no-cache/bypass.

11. Проверка

```

# На origin
curl -I https://ORIGIN_DOMAIN/
curl -i https://ORIGIN_DOMAIN/health
curl -i "https://ORIGIN_DOMAIN/api/v4/media/session/poll?offset=999" -H "Cookie: media_sid=debug" -H "X-Playback-Token: ping"

# Через CDN
curl -vk "https://CDN_TECH_DOMAIN/api/v4/media/session/poll?offset=999" -H "Cookie: media_sid=debug" -H "X-Playback-Token: ping"

# Логи
tail -f /var/log/nginx/xhttp_access.log /var/log/nginx/error.log

# Порты
ss -lntp | grep -E ':(80|443|11443)\b'

```

```
docker logs --tail=100 remnanode | grep -Ei 'cdn-get-inbound|get|usersCount|Xray started|error|failed' || true
```

Ожидаемые логи при пинге Happ:

```
GET /api/v4/media/session/poll/?offset=0 st=200
GET /api/v4/media/session/poll/?offset=1 st=200
GET /api/v4/media/session/poll/?offset=2 st=200
GET /api/v4/media/session/poll/ st=200 bytes=4150+
```

12. Fixed IP CDN edge

Для снижения джиттера можно создать отдельный Host с фиксированным IP edge. Address = IP, но SNI/Host остаются доменом CDN.

```
# PowerShell: собрать IP
$cdn = "CDN_TECH_DOMAIN"
$resolvers = @("1.1.1.1", "1.0.0.1", "8.8.8.8", "8.8.4.4", "9.9.9.9", "77.88.8.8")
$ips = @()
foreach ($resolver in $resolver) {
    1..20 | ForEach-Object {
        try { $r = Resolve-DnsName $cdn -Type A -Server $resolver -DnsOnly -ErrorAction Stop; $ips += $r.IPAddress } catch {}
        Start-Sleep -Milliseconds 150
    }
}
$ips | Where-Object { $_ } | Sort-Object -Unique

# PowerShell: замерить avg/min/max/spread
$cdn = "CDN_TECH_DOMAIN"
$ips = @("IP_1", "IP_2", "IP_3")
foreach ($ip in $ips) {
    $results = @()
    1..15 | ForEach-Object {
        $out = curl.exe -s -o NUL -w "%{time_total}" --resolve "${cdn}:443:${ip}" "https://${cdn}/health"
        if ($out) { $results += [double]$out }
        Start-Sleep -Milliseconds 300
    }
    $avg = ($results | Measure-Object -Average).Average
    $min = ($results | Measure-Object -Minimum).Minimum
    $max = ($results | Measure-Object -Maximum).Maximum
    $spread = $max - $min
    "{0} avg={1:N3}s min={2:N3}s max={3:N3}s spread={4:N3}s" -f $ip, $avg, $min, $max, $spread
}
```

Параметр	Значение
Fixed Host Address	Лучший IP edge
SNI	CDN_TECH_DOMAIN
Host	CDN_TECH_DOMAIN
Path	/api/v4/media/session/poll
Extra	тот же GET-only extra

13. Лимиты и мягкая очистка

Nginx и Docker limits

```
mkdir -p /etc/systemd/system/nginx.service.d
cat >/etc/systemd/system/nginx.service.d/limits.conf <<'EOF'
[Service]
LimitNOFILE=1048576
TasksMax=infinity
EOF

systemctl daemon-reload
nginx -t && systemctl restart nginx
```

```
NGINX_PID=$(pidof nginx | awk '{print $1}')
cat /proc/$NGINX_PID/limits | grep -i "open files"
```

```
docker exec remnanode sh -lc 'ulimit -n' || true
```

Мягкая очистка Nginx раз в сутки

```
cat >/usr/local/sbin/vpn-soft-cleanup.sh <<'EOF'
#!/usr/bin/env bash
set -euo pipefail

LOG="/var/log/vpn-soft-cleanup.log"

{
  echo "===== $(date -Is) ====="
  echo "[before] ss summary:"; ss -s || true
  echo "[before] 443 connections:"; ss -Htan sport = :443 | wc -l || true
  echo "[before] xray local 11443 connections:"; ss -Htan sport = :11443 | wc -l || true
  echo "[before] 443 states:"; ss -Htan sport = :443 | awk '{print $1}' | sort | uniq -c | sort -nr || true
  echo "[before] xray local 11443 states:"; ss -Htan sport = :11443 | awk '{print $1}' | sort | uniq -c | sort -nr || true

  if [ -f /proc/sys/net/netfilter/nf_conntrack_count ]; then
    echo "[before] conntrack:"
    echo -n "count: "; cat /proc/sys/net/netfilter/nf_conntrack_count
    echo -n "max: "; cat /proc/sys/net/netfilter/nf_conntrack_max
  fi

  echo "[action] nginx config test"; nginx -t
  echo "[action] restart nginx"; systemctl restart nginx
  sleep 2
  echo "[after] nginx status:"; systemctl is-active nginx || true
  echo "[after] ss summary:"; ss -s || true
  echo "[after] 443 connections:"; ss -Htan sport = :443 | wc -l || true
  echo "[after] xray local 11443 connections:"; ss -Htan sport = :11443 | wc -l || true
  echo
} >> "$LOG" 2>&1
EOF

chmod +x /usr/local/sbin/vpn-soft-cleanup.sh

cat >/etc/systemd/system/vpn-soft-cleanup.service <<'EOF'
[Unit]
Description=Soft cleanup for VPN CDN Nginx connections
After=network-online.target nginx.service
Wants=network-online.target

[Service]
Type=oneshot
ExecStart=/usr/local/sbin/vpn-soft-cleanup.sh
EOF

cat >/etc/systemd/system/vpn-soft-cleanup.timer <<'EOF'
[Unit]
Description=Run VPN soft cleanup daily

[Timer]
OnCalendar=*-*-* 04:10:00
Persistent=true
RandomizedDelaySec=120

[Install]
WantedBy=timers.target
EOF

systemctl daemon-reload
systemctl enable --now vpn-soft-cleanup.timer
systemctl list-timers --all | grep vpn-soft-cleanup
```

Команда статуса

```

cat >/usr/local/sbin/vpn-conn-status.sh <<'EOF'
#!/usr/bin/env bash

echo "==== $(date -Is) ====="
echo "--- ss summary ---"; ss -s

echo "--- 443 states ---"
ss -Htan sport = :443 | awk '{print $1}' | sort | uniq -c | sort -nr

echo "--- xray local 11443 states ---"
ss -Htan sport = :11443 | awk '{print $1}' | sort | uniq -c | sort -nr

echo "--- top remote IPs to 443, excluding LISTEN ---"
ss -Htan sport = :443 | awk '$1!="LISTEN"{print $5}' | sed 's/^\[/; s/\]$//' | cut -d: -f1 | sort | uniq -c | sort -nr | head -20

echo "--- conntrack ---"
if [ -f /proc/sys/net/netfilter/nf_conntrack_count ]; then
    echo -n "count: "; cat /proc/sys/net/netfilter/nf_conntrack_count
    echo -n "max: "; cat /proc/sys/net/netfilter/nf_conntrack_max
else
    echo "no conntrack"
fi

echo "--- file handles ---"; cat /proc/sys/fs/file-nr

echo "--- nginx timer ---"
systemctl list-timers --all | grep vpn-soft-cleanup || true
EOF
chmod +x /usr/local/sbin/vpn-conn-status.sh

```

14. Troubleshooting

Параметр	Значение
Логов нет при ping	Нapp/CDN не дошёл до origin: Address/SNI/Host/CDN/DNS/TLS.
400	Host extra и server extra не совпадают.
502	Nginx не достучался до 127.0.0.1:11443.
404 на curl	Нормально для невалидного xHTTP curl. Важно, что location логируется.
405	CDN режет метод. Для GET-only это не критично, POST/PUT выключить.
Высокий jitter	Пробуй fixed IP edge, более лёгкий xmux/padding, отключи HTTP/3/cache/gzip.
/ отдаёт Xray 400	Nginx всё ещё проксирует весь / в Xray. Перепиши site config.
Старые _dc / X-Cache в ответах	В Config Profile или Host extra остался старый padding header/key.

15. Финальный чек-лист

- ☐ ORIGIN_DOMAIN A -> ORIGIN_IP.
- ☐ Certbot выпустил сертификат для origin.
- ☐ Nginx / отдаёт HTML-заглушку, /health отдаёт JSON.
- ☐ Xray слушает 127.0.0.1:11443, не 0.0.0.0.
- ☐ Nginx proxy only /api/v4/media/session/poll -> 127.0.0.1:11443.
- ☐ Remnawave Config Profile и Host extra полностью совпадают.

- ☐ Host extra не содержит path.
- ☐ CDN cache/gzip/http3/secure token выключены для первого запуска.
- ☐ POST/PUT можно выключить.
- ☐ При ping Napp в xhttp_access.log идут GET offset=0/1/2/3 со статусом 200.
- ☐ Установлены лимиты и soft cleanup timer.